YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCES
TEST 2 - ALTERNATIVE SECTION 2

NAME: ____________________________ MARK: $\frac{\quad}{21}$

1. Marc Marquez accelerates at 13.2 m/s^2 from the start line in a Moto GP race in Spain. How long does it take him to reach 260 km/h if he is moving in a straight line towards the first corner? (3 marks)

2. A fully-loaded A-380 aeroplane weighing 560,000 kg lands at Sydney airport at 250km/h. It takes 17.5 s to slow to 60 km/h before leaving the runway and entering the taxi-way.

   (a) What deceleration was experienced by the passengers, assuming the braking force was uniform? (3 marks)

(b) What was the average force exerted by the brakes as it slowed the plane? (3 marks)

3. A car of mass 2100 kg accelerated from 90.0 km/h by the engine supplying a force of 2550 N. The frictional forces acting on the car totalled 430 N.

(a) What is the nett force acting on the car? (2 marks)

(b) What is the velocity of the car after 9.0 s of acceleration? (5 marks)

4. An object has a mass of 40.0 kg.

(a) What is the weight on Mars where the acceleration due to gravity is 3.71 m/s^2.
(2 marks)

(b) Would the object weigh more or less on the Earth? Explain your answer.
(2 marks)

(c) What is its mass on Mars? (1 mark)